

5.9.1

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

Construct a pair of tangents to a circle of radius 4 cm from a point P at a distance of 7 cm from its centre O .

Solution:

Given:

Let the centre be origin and P lies on x axis.

$$\mathbf{x}^T \mathbf{x} = 16 \quad (1)$$

$$\mathbf{P} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}, \quad (2)$$

Equation of a line through $\mathbf{P}$ with direction $\mathbf{d}$:

$$\mathbf{x} = \mathbf{P} + t\mathbf{d}, \quad t \in \mathbb{R}. \quad (3)$$

$$\text{Substitute (3) in (1) : } (\mathbf{P} + t\mathbf{d})^T (\mathbf{P} + t\mathbf{d}) = 16$$

$$\Rightarrow (\mathbf{d}^T \mathbf{d})t^2 + 2(\mathbf{P}^T \mathbf{d})t + (\mathbf{P}^T \mathbf{P} - 16) = 0 \quad (4)$$

For tangency, discriminant = 0:

$$\begin{aligned} (2\mathbf{P}^T \mathbf{d})^2 - 4(\mathbf{d}^T \mathbf{d})(\mathbf{P}^T \mathbf{P} - 16) &= 0 \\ \Rightarrow (\mathbf{P}^T \mathbf{d})^2 &= (\mathbf{d}^T \mathbf{d})(\mathbf{P}^T \mathbf{P} - 16) \end{aligned} \quad (5)$$

$$\mathbf{P}^T \mathbf{P} = 49, \quad (6)$$

$$(\mathbf{P}^T \mathbf{d})^2 = 33(\mathbf{d}^T \mathbf{d}). \quad (7)$$

For a unit vector $\mathbf{d}$,

$$\mathbf{d}^T \mathbf{d} = 1 \Rightarrow (\mathbf{P}^T \mathbf{d})^2 = 33 \quad (8)$$

Let

$$\mathbf{M} = \mathbf{P}\mathbf{P}^T \implies \mathbf{d}^T \mathbf{M} \mathbf{d} = (\mathbf{P}^T \mathbf{d})^2 \quad (9)$$

$$\mathbf{d}^T \mathbf{M} \mathbf{d} = 33, \quad (10)$$

$$\mathbf{M} = \mathbf{P}\mathbf{P}^T = \begin{pmatrix} 7 \\ 0 \end{pmatrix} \begin{pmatrix} 7 & 0 \end{pmatrix} = \begin{pmatrix} 49 & 0 \\ 0 & 0 \end{pmatrix} \quad (11)$$

Eigen values and Eigen vectors of M:

$$\lambda_1 = 49, \lambda_2 = 0.$$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (12)$$

$\mathbf{d}$ in orthonormal Eigenbasis : $\mathbf{d}^T \mathbf{d} = 1$

$$\mathbf{d} = \alpha \mathbf{e}_1 + \beta \mathbf{e}_2, \quad \alpha^2 + \beta^2 = 1 \quad (13)$$

$$\mathbf{d}^T \mathbf{M} \mathbf{d} = \lambda_1 \alpha^2 + \lambda_2 \beta^2 \quad (14)$$

$$\implies \mathbf{d}^T \mathbf{M} \mathbf{d} = 49\alpha^2 + 0 \cdot \beta^2 = 49\alpha^2 \quad (15)$$

$$\text{From (10), } 49\alpha^2 = 33 \implies \alpha = \pm \frac{\sqrt{33}}{7}. \quad (16)$$

$$\implies \beta = \pm \frac{4}{7} \quad (17)$$

Hence $\mathbf{d}$ is:

$$\therefore \mathbf{d} = \begin{pmatrix} \frac{\sqrt{33}}{7} \\ \pm \frac{4}{7} \end{pmatrix} \quad (18)$$

From (4) and (8)

$$t = -\frac{\mathbf{P}^T \mathbf{d}}{\mathbf{d}^T \mathbf{d}}. \\ \implies t = -\sqrt{33} \quad (19)$$

Required Points of Contact

$$\mathbf{T} = \mathbf{P} + t\mathbf{d} = \mathbf{P} - \sqrt{33}\mathbf{d} \quad (20)$$

$$\mathbf{P} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}, \mathbf{d} = \begin{pmatrix} \frac{\sqrt{33}}{7} \\ \pm \frac{4}{7} \end{pmatrix} : \quad (21)$$

$$\mathbf{T} = \begin{pmatrix} 7 \\ 0 \end{pmatrix} - \sqrt{33} \begin{pmatrix} \frac{\sqrt{33}}{7} \\ \pm \frac{4}{7} \end{pmatrix} = \begin{pmatrix} \frac{16}{7} \\ \mp \frac{4\sqrt{33}}{7} \end{pmatrix} \quad (22)$$

$$\mathbf{T}_1 = \begin{pmatrix} \frac{16}{7} \\ \frac{4\sqrt{33}}{7} \end{pmatrix}, \quad \mathbf{T}_2 = \begin{pmatrix} \frac{16}{7} \\ -\frac{4\sqrt{33}}{7} \end{pmatrix} \quad (23)$$

Equation of tangents :

$$\mathbf{x} = \begin{pmatrix} 7 & 0 \end{pmatrix}^T + k \begin{pmatrix} \frac{\sqrt{33}}{7} & \frac{4}{7} \end{pmatrix}^T \quad (24)$$

$$\mathbf{x} = \begin{pmatrix} 7 & 0 \end{pmatrix}^T + k \begin{pmatrix} \frac{\sqrt{33}}{7} & -\frac{4}{7} \end{pmatrix}^T \quad (25)$$

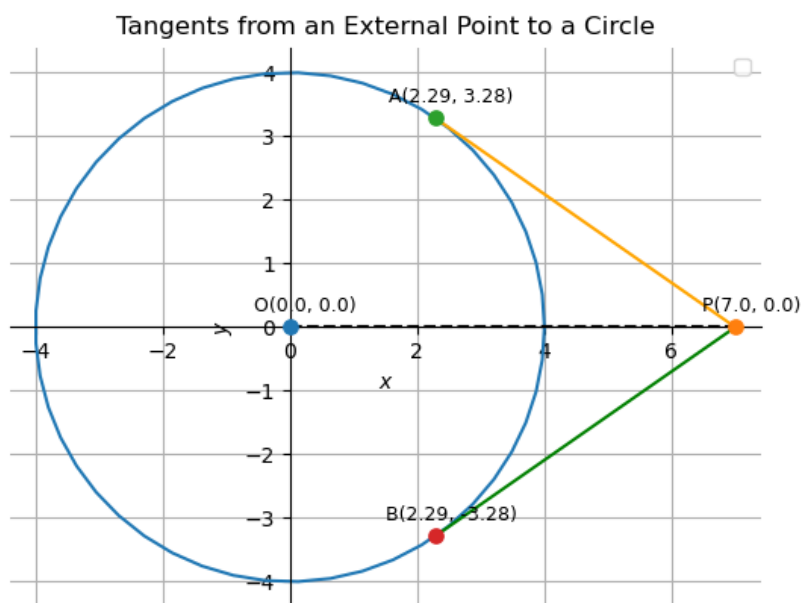


Figure 1